AI BASED INDOOR ROBOT NAVIGATION USING A* SEARCH AND DYNAMIC OBSTACLE AVOIDANCE

CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

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IN

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

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INTERNAL EXAMINER EXTERNAL EXAMINER

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ABSTRACT

Indoor robot navigation is an important application of Artificial Intelligence and robotics, especially in environments such as hospitals, offices, warehouses, laboratories, shopping centres, and homes. A major challenge in indoor navigation is enabling a robot to reach a destination efficiently while avoiding obstacles that may change their position.

This project proposes an AI Based Indoor Robot Navigation System using A Search and Dynamic Obstacle Avoidance*. The system represents the indoor environment as a grid-based map containing free spaces and obstacles. The A* search algorithm is used to calculate an efficient path between the robot's current position and the target destination. A* combines the actual movement cost with a heuristic estimate of the remaining distance, allowing the robot to select a suitable path.

However, a static path alone is not sufficient for real indoor environments because people, objects, and other robots may move into the planned path. Therefore, a dynamic obstacle avoidance mechanism is incorporated into the system. Sensors or simulated sensor data are used to identify obstacles appearing along the robot's route. When a dynamic obstacle is detected, the system evaluates the current path and determines whether the robot can safely continue, wait, or calculate a new path.

The proposed system follows a systematic process involving environment representation, obstacle detection, path planning, path execution, dynamic obstacle monitoring, and path replanning. The system is evaluated using different indoor maps and obstacle conditions.

The project demonstrates how Artificial Intelligence algorithms can be combined with robotic sensing and decision-making to achieve safer and more efficient indoor navigation. It also provides a practical understanding of path-search algorithms, heuristic functions, obstacle avoidance, simulation, and autonomous robot behaviour.

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CHAPTER 1

INTRODUCTION

1.1 Background Information

With the advancement of Artificial Intelligence, robotics has become an important area for developing autonomous systems capable of sensing their environment and making decisions without continuous human control.

Indoor robot navigation is one of the important problems in autonomous robotics. A robot operating inside a building must determine its current location, identify obstacles, select a suitable route, and reach a specified destination safely.

Traditional navigation methods may work well in environments where obstacles remain fixed. However, indoor environments are usually dynamic. People may walk across corridors, objects may be moved, doors may be opened or closed, and other robots may enter the robot's path.

The proposed project addresses this problem using *A Search** for path planning and a **Dynamic Obstacle Avoidance** mechanism for responding to changes in the environment.

A* is a graph-search algorithm that evaluates possible paths using:

f(n) = g(n) + h(n)

where:

- **g(n)** represents the actual cost from the starting position to the current node.
- **h(n)** represents the estimated cost from the current node to the destination.
- **f(n)** represents the total estimated path cost.

The robot initially creates a path using A*. During movement, the environment is continuously monitored. If a dynamic obstacle blocks the planned route, the robot identifies the obstruction and recalculates an alternative route.

This combination allows the system to demonstrate intelligent and adaptive indoor navigation.

1.2 Project Objectives

The main objectives of this capstone project are to:

- Design an AI-based indoor robot navigation system.
- Represent an indoor environment using a grid-based map.
- Implement the A* search algorithm for path planning.

- Calculate an efficient route between source and destination.
- Detect obstacles appearing along the robot's path.
- Implement dynamic obstacle avoidance.
- Recalculate the path when the original route becomes blocked.
- Reduce unnecessary movement and navigation errors.
- Test the system under different indoor obstacle conditions.
- Evaluate the effectiveness of the proposed navigation approach.

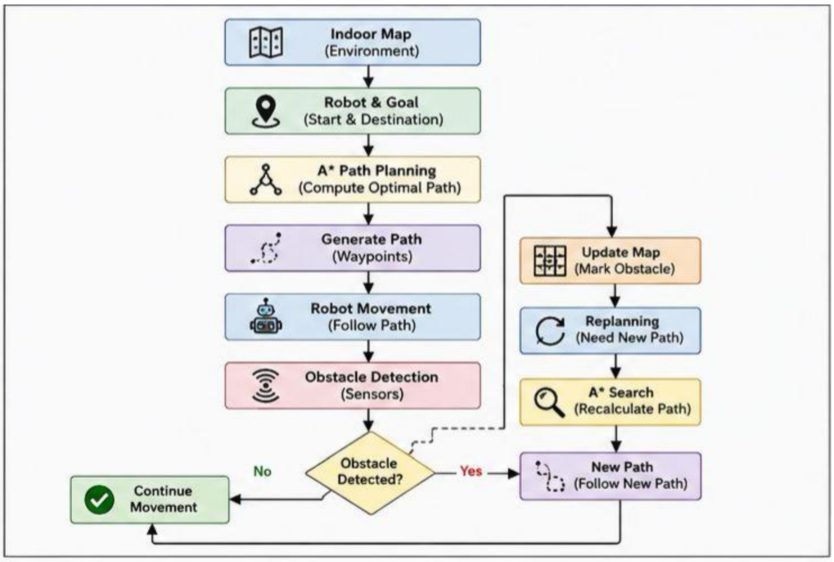

Fig 1 : System Architecture

1.3 Significance of the Project

The proposed project is significant because it demonstrates how Artificial Intelligence can be applied to autonomous robotic navigation.

The system:

- Reduces dependence on manual robot control.
- Provides efficient path planning using A*.
- Allows the robot to respond to changing environments.
- Improves navigation safety by identifying obstacles.
- Demonstrates real-time decision-making.
- Reduces the possibility of collision.
- Provides a foundation for autonomous indoor robots.

- Helps connect theoretical AI algorithms with practical robotics applications.

The system can be useful in indoor environments such as hospitals, warehouses, offices, laboratories, and educational institutions.

1.4 Scope of the Project

In-Scope

- Indoor grid-map representation.
- Source and destination selection.
- Static obstacle representation.
- A* path planning.
- Dynamic obstacle detection.
- Path replanning.
- Robot movement simulation.
- Performance evaluation using different test cases.

Out-of-Scope

- Outdoor GPS-based navigation.
- Complex autonomous vehicle navigation.
- Large-scale industrial robot deployment.
- Advanced simultaneous localization and mapping systems.
- Full commercial-grade hardware implementation.
- Multi-robot coordination.

1.5 Methodology Overview

The project follows a systematic development approach:

Research and Analysis

Study indoor navigation, A* search, heuristic functions, obstacle detection, and dynamic path planning.

Design

Design the indoor map, robot representation, obstacle representation, path-planning module, and dynamic obstacle detection module.

Implementation

Implement the A* algorithm and obstacle avoidance mechanism using suitable programming tools.

Testing

Test the system using different maps, starting points, destinations, static obstacles, and dynamic obstacles.

Evaluation

Evaluate path efficiency, navigation success, obstacle response, replanning, and overall system performance.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Indoor robots must navigate through environments containing walls, furniture, people, and other obstacles. The robot needs to reach its destination without colliding with these objects.

A simple fixed path may not be suitable because the environment can change after the path has been generated.

For example, consider a robot travelling through a corridor. The A* algorithm may initially generate a path through the corridor. If a person suddenly enters the corridor, the robot must recognize that the path is no longer safe.

Therefore, the system must perform two major tasks:

1. **Path Planning** – Find a suitable path from the robot to the destination.
2. **Dynamic Obstacle Avoidance** – Detect changes and modify the path when necessary.

The project combines these two capabilities to provide adaptive indoor navigation.

2.2 Evidence of the Problem

Several practical situations demonstrate the need for intelligent indoor navigation:

- People continuously move inside buildings.
- Furniture and objects may change position.
- Corridors may become temporarily blocked.
- Static paths cannot respond automatically to environmental changes.
- Manual navigation requires continuous human supervision.
- A robot following an outdated path may collide with an obstacle.
- Recalculating the route can improve navigation reliability.

Therefore, an indoor robot needs both path planning and obstacle-response capabilities.

2.3 Stakeholders

The following groups can benefit from an intelligent indoor navigation system:

Robot Developers

Use the system to develop and test autonomous navigation algorithms.

Researchers

Can use the project as a foundation for studying path planning and obstacle avoidance.

Hospitals

Autonomous robots can potentially transport medicines, documents, or supplies.

Warehouses

Robots can navigate between storage areas while responding to moving workers and objects.

Offices

Robots can be used for delivery and assistance tasks.

Educational Institutions

The project can be used as a practical demonstration of AI, robotics, and search algorithms.

2.4 Supporting Data / Research

Research in Artificial Intelligence and robotics has established search-based path planning as an important approach for autonomous navigation.

The A* algorithm is widely studied for finding paths in graphs and grid environments. Its combination of path cost and heuristic estimation allows it to search efficiently toward a target.Dynamic environments require additional mechanisms because a previously calculated path can become invalid when obstacles move.

The proposed project therefore combines:

- Grid-based environment modelling.
- Heuristic search.
- A* path planning.
- Obstacle detection.
- Dynamic path replanning.
- Robot movement simulation.

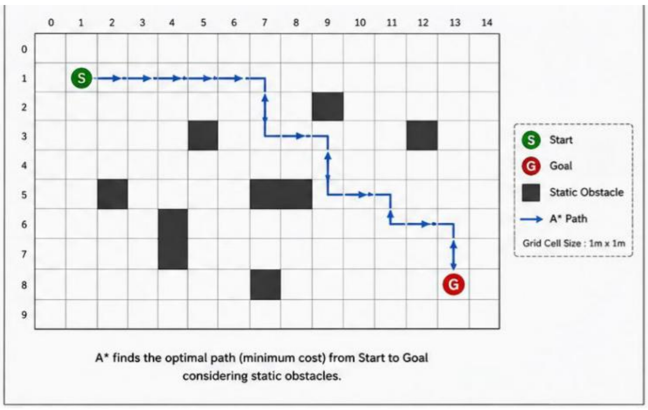

Fig 2 :A* path planning

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development of the proposed system follows a structured software engineering process involving requirement analysis, system design, implementation, testing, and evaluation.

The system is divided into several modules:

Environment Module

Represents the indoor environment as a grid.

Robot Module

Stores the robot's current position and movement information.

Destination Module

Defines the target location that the robot needs to reach.

A* Path Planning Module

Calculates the path from the current position to the destination.

Obstacle Detection Module

Identifies static and dynamic obstacles.

Dynamic Avoidance Module

Determines whether the robot can continue along its current route.

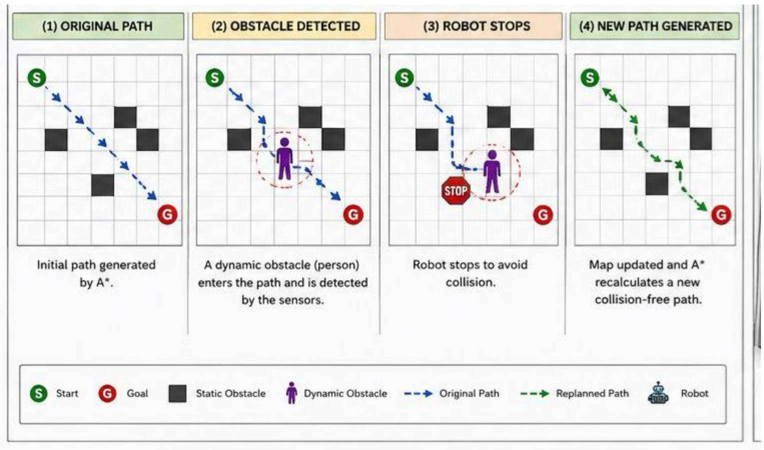

Fig 3 :Dynamic Avoidance

Replanning Module

Generates a new path when the original path becomes blocked.

Visualization Module

Displays the map, robot, obstacles, and calculated path.

3.2 Tools and Technologies Used

Programming Language

Python can be used to implement the navigation algorithm because it provides simple syntax and useful libraries for simulation and visualization.

Algorithms

- A* Search Algorithm
- Heuristic-based path planning
- Dynamic obstacle detection
- Path replanning

Data Structures

- Grid
- Priority Queue
- Open List
- Closed List
- Parent/Predecessor Map
- Cost Map

Visualization

A graphical visualization can be used to display:

- Robot position.
- Target position.
- Static obstacles.
- Dynamic obstacles.
- Planned path.
- Replanned path.

Hardware Possibilities

For a physical implementation, the system can be integrated with:

- Ultrasonic sensors.
- Infrared sensors.
- LiDAR.
- Wheel motors.
- Motor drivers.
- Microcontroller or single-board computer.

S.No.	Component	Technology / Method Used	Purpose
1	Programming Language	Java	Implementation of the navigation system
2	Path Planning Algorithm	A* Search Algorithm	Finds an efficient path from start to goal
3	Environment Representation	Grid-Based Map	Represents free spaces and obstacles
4	Data Structures	Priority Queue, Lists, Maps	Stores nodes, costs, and path information
5	Obstacle Detection	Simulated Sensor Data	Detects dynamic obstacles
6	Path Replanning	A* Search	Generates a new path when blocked
7	Visualization	Java Swing / JavaFX	Displays robot movement and navigation

Table 1 :Tools and Technologies Used

3.3 Solution Overview

The proposed system operates through the following stages:

1. Create Indoor Map

The indoor environment is represented using a grid.

Example:

S . . # . . .

. . . # . . .

. # . . . # .

. . . .

. . # . . . G

Where:

- **S** = Starting position
- **G** = Goal
- **#** = Obstacle
- **.** = Free space

2. Select Start and Goal

The robot's current position is defined as the starting point, and the required destination is selected as the goal.

3. Apply A* Search

The A* algorithm evaluates available grid cells and calculates:

f(n) = g(n) + h(n)

The algorithm continues until the goal is reached or no valid path exists.

4. Generate Path

The system reconstructs the path from the goal back to the starting position using the stored parent nodes.

5. Start Robot Movement

The robot follows the generated path step-by-step.

6. Monitor Environment

During movement, the system continuously checks for new obstacles.

7. Detect Dynamic Obstacle

If an obstacle appears on the planned route, the robot stops or slows down depending on the situation.

8. Recalculate Path

The affected grid cell is marked as blocked and A* is executed again from the robot's current position.

9. Continue Navigation

The robot follows the newly generated route until it reaches the destination.

3.4 Engineering Standards Applied

The project can follow recognized engineering and coding practices.

Software Development Lifecycle

The project follows structured phases:

- Requirement analysis.
- System design.
- Implementation.
- Testing.
- Evaluation.
- Maintenance and improvement.

Coding Standards

Python coding can follow **PEP 8** principles to improve readability and maintainability.

Testing Standards

Test cases should be systematically documented for:

- Valid navigation.
- Blocked routes.
- Dynamic obstacles.
- No available route.
- Start and goal positions.
- Multiple obstacles.

Modular Design

The system is divided into independent modules for:

- Map generation.
- Path planning.
- Obstacle detection.

- Path replanning.
- Visualization.

3.5 Solution Justification

A* search is selected because it provides an effective method for finding a path between two locations in a grid environment.

The use of a heuristic allows the search to focus toward the destination instead of exploring the entire environment unnecessarily.

However, A* alone cannot solve the complete problem because the environment can change after the path has been generated.

Therefore, dynamic obstacle avoidance is added to:

- Monitor the environment.
- Identify changes.
- Prevent collision.
- Trigger path replanning.
- Maintain navigation toward the destination.

The combination of A* and dynamic obstacle avoidance provides a practical approach to indoor autonomous navigation.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The proposed system is evaluated using different indoor navigation scenarios.

Accuracy

The system successfully generates a path when a valid route exists. It avoids cells marked as obstacles during path planning.

Dynamic Obstacle Response

When an obstacle is introduced into the robot's planned path, the system identifies the blocked section and initiates path replanning.

Path Efficiency

The A* algorithm attempts to generate an efficient route based on movement cost and heuristic estimation.

Navigation Success

The robot successfully reaches the destination when an alternative path is available.

Failure Handling

If all possible routes are blocked, the system reports that no valid path is currently available.

Adaptability

The dynamic obstacle mechanism allows the system to respond to changes rather than continuously following an outdated path.

4.2 Challenges Encountered

Several challenges may be encountered during development:

Dynamic Obstacle Detection

Detecting moving obstacles correctly can be difficult because their positions change continuously.

Path Replanning

Repeatedly calculating a new route can increase computational requirements.

Sensor Limitations

Physical sensors may produce inaccurate or noisy measurements.

Narrow Spaces

Navigation becomes difficult when the robot has limited space to move around an obstacle.

Multiple Obstacles

A large number of obstacles can increase the complexity of path planning.

Real-Time Processing

The system must detect environmental changes and respond quickly enough to avoid collisions.

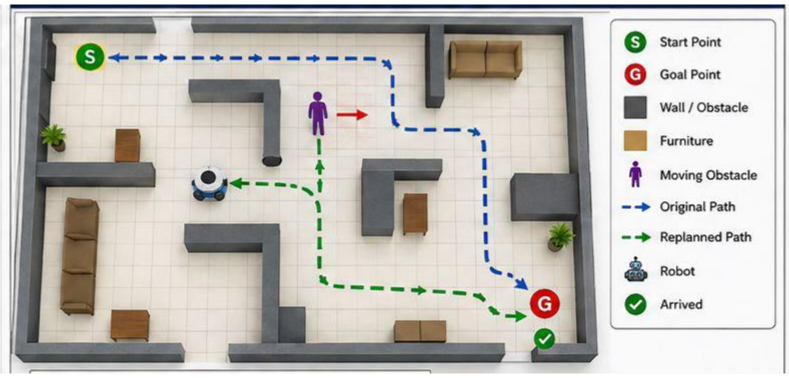

Fig 4 :Real-Time Processing

4.3 Possible Improvements

The project can be improved by:

- Integrating LiDAR for accurate environment sensing.
- Adding real-time camera-based object detection.
- Using machine learning for obstacle classification.
- Implementing adaptive speed control.
- Adding robot localization.
- Supporting larger indoor maps.
- Using more advanced dynamic path-planning algorithms.
- Supporting multiple robots.
- Adding real-time visualization.
- Integrating the system with a physical mobile robot.

4.4 Recommendations

Based on the project design, the following recommendations are proposed:

- Test the system using different indoor layouts.
- Use real sensor data for physical implementation.
- Measure path length and navigation time.
- Test different obstacle speeds.
- Include safety distance around obstacles.
- Evaluate performance under crowded conditions.
- Compare A* with other navigation algorithms.
- Improve the system's real-time response.
- Conduct extensive hardware testing before real-world deployment.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Through this project, several important technical and practical skills were developed.

Understanding of Artificial Intelligence

The project provided an understanding of how AI algorithms can be used to solve navigation problems.

Understanding of A* Search

The project improved knowledge of:

- Search algorithms.
- Heuristic functions.
- Path cost.
- Open and closed lists.
- Path reconstruction.

Robotics Knowledge

The project provided an understanding of robot movement, sensing, obstacle detection, and autonomous decision-making.

Programming Skills

Implementation improved programming skills related to:

- Data structures.
- Algorithms.
- Functions.
- Grid processing.
- Visualization.
- Testing.

Problem-Solving Skills

The project required analysing navigation problems and developing solutions for changing environments.

5.2 Challenges Encountered and Overcome

Challenge: Finding an Efficient Path

Initially, selecting a suitable route through obstacles can be difficult.

Solution: The A* algorithm was implemented using movement cost and heuristic estimation.

Challenge: Dynamic Obstacles

A path generated at the beginning may become blocked later.

Solution: A dynamic obstacle monitoring mechanism was incorporated to trigger path replanning.

Challenge: No Available Route

Some maps may not contain a valid route to the destination.

Solution: The system was designed to identify when no path exists and report the condition.

Challenge: Real-Time Response

The robot must respond quickly when an obstacle appears.

Solution: The navigation process was divided into path planning and continuous obstacle monitoring modules.

5.3 Application of Engineering Standards

The project applied engineering principles through:

- Modular software design.
- Structured testing.
- Algorithm analysis.
- Documentation.
- Code readability.
- Systematic development.
- Performance evaluation.

Separating path planning, obstacle detection, and visualization improves maintainability and makes future modifications easier.

5.4 Insights into the Industry

The project provided an understanding of how autonomous navigation can be applied to real-world robotics.

Indoor autonomous robots can potentially be used for:

- Hospital delivery.
- Warehouse transportation.
- Office assistance.
- Security monitoring.
- Laboratory automation.
- Service robotics.

The project also demonstrated that successful autonomous navigation requires more than a path-planning algorithm. The robot must continuously sense and respond to its environment.

5.5 Conclusion of Personal Development

This project improved technical knowledge, algorithmic thinking, programming ability, and understanding of autonomous systems.

It helped connect theoretical Artificial Intelligence concepts with practical robotics applications.

The project also improved the ability to analyse problems, design modular solutions, conduct testing, and evaluate system performance.

CHAPTER 6

CONCLUSION

The project “AI Based Indoor Robot Navigation Using A Search and Dynamic Obstacle Avoidance”* demonstrates an intelligent approach for navigating robots through indoor environments.

The system uses a grid-based representation of the environment and applies the A* search algorithm to generate an efficient path between the robot and its destination. The calculated route enables the robot to navigate around known obstacles.

Since real indoor environments can change, a dynamic obstacle avoidance mechanism is incorporated. When an obstacle appears along the planned route, the system identifies the blockage and recalculates an alternative path from the robot's current position.

The project demonstrates the importance of combining path planning with environmental awareness. A* provides effective route planning, while dynamic obstacle avoidance enables the robot to respond to changing conditions.

The proposed system can serve as a foundation for future autonomous indoor robots used in hospitals, warehouses, offices, laboratories, and other indoor environments.

Future development can include real-time camera processing, LiDAR-based mapping, advanced localization, machine-learning-based obstacle recognition, multi-robot coordination, and implementation on a physical robot.

In conclusion, the project successfully demonstrates how Artificial Intelligence, search algorithms, and obstacle avoidance techniques can be integrated to create an adaptive indoor robot navigation system.

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APPENDICES

APPENDIX A

Sample Indoor Grid Map

S . . . # . . . .

. . # . # . # . .

. . # . . . # . .

. . . .

. # . # . . .

. . . .

. # . . . . # . .

. . . # . . . . G

Where:

- **S** = Robot starting position
- **G** = Goal position
- **#** = Obstacle
- **.** = Free space

APPENDIX B

A* Path Planning Procedure

1. Create the indoor grid.
2. Set the robot position as Start.
3. Set the destination as Goal.
4. Add the Start node to the Open List.
5. Select the node with the lowest f(n).
6. Check its neighbouring cells.
7. Ignore blocked cells.
8. Calculate g(n), h(n), and f(n).
9. Add valid neighbouring nodes to the Open List.

10. Move the processed node to the Closed List.

11. Repeat until Goal is reached.

12. Reconstruct the path using parent nodes.

13. Start robot movement.

14. Monitor for dynamic obstacles.

15. If the path is blocked, update the map.

16. Run A* again from the current robot position.

17. Follow the new path.

18. Stop when the robot reaches the Goal.

APPENDIX C

Sample Test Cases

Test Case	**Condition**	**Expected Result**
TC01	Empty indoor map	Robot reaches destination
TC02	Static obstacles	A* finds a path around obstacles
TC03	Dynamic obstacle appears	Robot detects obstacle
TC04	Obstacle blocks current path	Robot calculates a new path
TC05	Multiple obstacles	Robot navigates through available space
TC06	Complete blockage	System reports no valid path
TC07	Goal reached	Navigation successfully terminates

APPENDIX D

Sample System Output

Indoor Robot Navigation System

Robot Start Position : (0,0)

Goal Position : (7,8)

Calculating path using A*...

Initial Path:

(0,0) → (0,1) → (1,1) → (2,1) → (3,1)

→ (3,2) → (3,3) → (4,3) → (5,3)

→ (6,4) → (7,5) → (7,6) → (7,7) → (7,8)

Robot Movement Started...

Dynamic obstacle detected at (5,3)

Current path blocked.

Stopping robot...

Recalculating path using A*...

New Path Generated.

Robot continuing movement...

Destination reached successfully.

Result: Navigation Successful

APPENDIX E

Pseudocode for Dynamic Navigation

```
procedure Robot_Navigation(Map, Start, Goal):
    CurrentPosition ← Start
    while CurrentPosition != Goal:
        Path ← A_Star(Map, CurrentPosition, Goal)
        if Path does not exist:
            return "No valid path"
        for each Position in Path:
            if Dynamic_Obstacle_Detected(Position):
                Stop Robot
                Mark Position as Blocked
```

```
        break
      else:
        Move Robot to Position
        CurrentPosition ← Position
  return "Destination Reached"
```

APPENDIX F

A* Cost Calculation

For every candidate node:

f(n) = g(n) + h(n)

Where:

- **g(n)** = cost from starting position to current node.
- **h(n)** = estimated cost from current node to destination.
- **f(n)** = total estimated cost.

For a grid-based indoor environment, a suitable heuristic can be selected according to the allowed movement model.

The node with the lowest estimated total cost is selected for further exploration.

APPENDIX G

Overall System Workflow

Start

↓

Create Indoor Map

↓

Set Robot Position

↓

Set Destination

↓

Detect Static Obstacles

↓

Apply A* Search

↓

Generate Optimal Path

↓

Start Robot Movement

↓

Monitor Environment

↓

Dynamic Obstacle Detected?

↓ Yes

Stop Robot

↓

Update Map

↓

Run A* Again

↓

Generate New Path

↓

Continue Movement

↓

Destination Reached?

↓ Yes

Stop

This structure follows the reference document's progression from background and objectives → problem analysis → solution design → implementation → evaluation → learning → conclusion → appendices, while replacing the PDA-specific material with content appropriate to your robot-navigation project.